



PROFILE

Civil Engineer with a Master's Degree in Structural Engineering. Passionate about structural optimization and adaptivity as key drivers for the next generation of sustainable buildings. Skilled in structural analysis, numerical modeling, and advanced simulation using FEM, BIM platforms, and programming languages (MATLAB, Python, C#). Strongly motivated to bridge technological innovation, architectural beauty and sustainability in the construction industry. Open-minded, proactive, and eager to thrive in dynamic and international working environments.

CONTACT

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HOBBY

Sports: tennis, skiing, running

Passions: motorsport, travel, DIY and model making

FILIPPO CASTRIOTA SCANDERBEG

CIVIL STRUCTURAL ENGINEER

EDUCATION

Master's Degree in Civil Engineering (Structures) - University of Florence
04/2022 – 04/2026 (Grade: 109/110)

- Thesis: Topology Optimization of Adaptive Structures - Mathematical Formulation Analysis and Method implementation for Floor Slab (Prof. Gianni Bartoli (UniFi), Dr. Gennaro Senatore (University of Stuttgart, ILEK), Prof. Luca Facchini (UniFi))
- Main courses: Dynamics of structures, Theory and design of structures, Complements of geotechnics, Numerical analysis and computational mechanics, Road infrastructure design, Theory and design of bridges, Computational analysis of structures, Fluid dynamics, Wind engineering, Earthquake Engineering.

Bachelor's Degree in Building Engineering - University of Florence
09/2018 – 04/2022 (Grade: 110/110)

- Thesis: Numerical Modeling of a Reinforced Concrete Building (Prof. Luca Facchini, Prof. Gianni Bartoli) - Grade: 110/110
- Main courses: Math I and II, Physics I and II, Architecture and architectural composition, Topography and GIS, Technical drawing, Chemistry and technology of materials, Geometry, BIM, Technical architecture, Solid mechanics, Building production and safety, Building Science, Hydraulics, Geotechnics, Science of innovative materials.

Niccolò Rodolico Scientific High School
2013 – 2018 (Grade: 95/100)

- Applied Sciences

PROFESSIONAL EXPERIENCE

University of Stuttgart, Institute for Lightweight Structures and Conceptual Design (ILEK) – Intern

03/2024 – 09/2024

Internship experience in design and fabrication of adaptive ribbed floor systems:

- Structural analysis and verification of a prototype floor slab (10×6 m) with adaptive post-tensioning system;
- Prototype design and fabrication;
- Study and in-depth analysis of structural optimization methods as material for the development of the master's thesis.

Studio Ing. Costantino Castriota Scanderbeg – Assistant

08/2022 – 09/2022

Supervision of the expansion of a reinforced concrete warehouse in San Polo in Chianti:

- Assistance in the reinforcement and casting phases of foundations and main structural elements.

PUBLICATIONS

- Senatore, G., Castriota Scanderbeg, F., Bartoli, G. (2026). "Topology Optimization of Passive-Active Continua Under Eigenstrain Actuation with a Controlled-Compliance Objective", abstract accepted for presentation at IASS-IWSS 2026, Turin, Italy, Sep. 14–18, 2026.

SKILLS

Technical skills

- Structural calculation / FEM: SAP2000, Midas Gen, MATLAB, Python, C#
- CAD / BIM: AutoCAD 2D, Revit
- Parametric modeling: Rhinoceros, Grasshopper
- Other: Office package

Languages

- Italian: Native speaker
- English: Cambridge First Certification – B2 (listening, reading, speaking)
- German: A1 (basic)